

# Technology

The technology layer — signalling, power, guideway and automation — and why Taipei runs several of each.

Taipei operates two incompatible medium-capacity systems, several signalling generations, rubber-tyred and steel-wheel running, and both third-rail and overhead power. Nothing in English explains why, and the answer is mostly history — procurement decisions whose consequences outlived the contracts that made them. The Matra dispute [\[/rail/history/matra-dispute/\]](#) is the opening chapter of that story; the technical threads — automation grades, control centres, depot maintenance tiers, electrification, fire and evacuation design, platform doors, seismic design, signalling generations, track gauge and station-numbering — each have their own page below.

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13 entries

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## Metro systems and operations

Signalling, traction power, depots, service spacing, first and last trains, and platform doors across Taipei-region metro lines.

### Three numbering systems on one line

 Wenhu Line

Every Wenhu station has two numbers. On the southern half they run in opposite directions, and the northern half uses a different series again — because the line was built as two projects and a station was added to one of them late.

## Metro depot maintenance levels

What Taipei's Level 1 through Level 5 depot labels actually describe, from pre-departure checks to network-wide overhaul.

## Neihu Line steel running surface

Why the Neihu half of the Wenhu Line uses a pre-bent, coated steel running surface while the older Muzha half uses reinforced concrete.

## Taipei Metro platform screen doors

How Taipei added platform screen doors to the early high-capacity network, and what the primary record does and does not say about completion by line.

### Why the Wenhu Line looks the way it does

 Wenhu Line

The official design logic behind the Wenhu Line's steel-and-glass stations, its three Neihu architectural vocabularies, and its river-in-the-sky idea.

## Taipei-region metro automation grades

What GoA0 through GoA4 mean on Taipei-region railways, and which current lines have a published automation grade.

## Taipei Metro signalling generations

The published boundary between Taipei's fixed-block high-capacity signalling and its later moving-block CBTC systems.

## Taipei-region metro electrification

Voltage, collection method and traction-substation boundaries across Taipei Metro, New Taipei Metro and the Airport MRT.

## Taipei-region metro control centres

What the three Taipei-region metro operators publish about their control-centre arrangements, and what remains undisclosed.

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Taipei-region metro track gauges

The published gauge boundary between Taipei's steel-wheel metro, the rubber-tyred Wenhua system and New Taipei light rail.

Taipei Metro seismic design boundaries

The sourced chronology from Taipei Metro's early seismic basis through structure-specific design methods and the current national-code comparison.

Taipei-region metro fire and evacuation systems

The published boundaries between fire detection, smoke control, life-safety systems, vehicle egress, fixed escape paths and the 2025 Circular Line emergency response.

Side by side

|                                                         | Metro systems and operations | Metro depot maintenance levels                       |
|---------------------------------------------------------|------------------------------|------------------------------------------------------|
| Taipei Metro service day                                | 06:00–24:00                  | —                                                    |
| Wenhua peak headway                                     | 2–4 min                      | —                                                    |
| Tamsui-Xinyi peak headway                               | 6 min, with 3 min overlap    | —                                                    |
| Songshan-Xindian peak headway                           | 4–6 min, with 3 min overlap  | —                                                    |
| Zhonghe-Xinlu peak headway                              | 6 min, with 3 min overlap    | —                                                    |
| Bannan peak headway                                     | 6 min, with 3 min overlap    | —                                                    |
| Level 3 additions                                       | —                            | Wheel reprofiling and annual route-train maintenance |
| Level 4 additions                                       | —                            | Emergency repairs, spares, non-routine inspection    |
| Level 5 additions                                       | —                            | Network-wide inspection, major overhaul and repair   |
| Detailed cycles                                         | —                            | Set by each operator and approved by the company     |
| Neihu section length in the paper                       | —                            | —                                                    |
| Reported section steel                                  | —                            | —                                                    |
| Reported fishplate quantity                             | —                            | —                                                    |
| High-capacity platform-door arrangement                 | —                            | —                                                    |
| High-capacity platform length                           | —                            | —                                                    |
| Xinyi and Songshan design                               | —                            | —                                                    |
| Neihu section                                           | —                            | —                                                    |
| Neihu underground running                               | —                            | —                                                    |
| Neihu underground stations                              | —                            | —                                                    |
| Neihu elevated bridge described in the design record    | —                            | —                                                    |
| GoA2 operating arrangement                              | —                            | —                                                    |
| GoA4 operating arrangement                              | —                            | —                                                    |
| GoA4 control family in the Circular and Sanying records | —                            | —                                                    |
| High-capacity separation method                         | —                            | —                                                    |
| Wenhua position/control basis                           | —                            | —                                                    |
| Circular location boundary                              | —                            | —                                                    |
| High-capacity reset programme value                     | —                            | —                                                    |
| High-capacity traction-substation spacing               | —                            | —                                                    |
| Medium-capacity traction-substation spacing             | —                            | —                                                    |
| Wenhua incoming supply                                  | —                            | —                                                    |
| Airport technical-page equipment count                  | —                            | —                                                    |
| TRTC centres in published 2014 arrangement              | —                            | —                                                    |
| TYMC named line control centres                         | —                            | —                                                    |

|                                                                   | Metro systems and operations | Metro depot maintenance levels |
|-------------------------------------------------------------------|------------------------------|--------------------------------|
| High-capacity design gauge                                        | —                            | —                              |
| New Taipei light-rail vehicle gauge                               | —                            | —                              |
| Wenhu guideway type                                               | —                            | —                              |
| Early metro maximum horizontal ground acceleration                | —                            | —                              |
| CEDM tenth-edition maximum horizontal ground acceleration         | —                            | —                              |
| Underground-station ODE                                           | —                            | —                              |
| Underground-station MDE                                           | —                            | —                              |
| Cut-and-cover tunnel ODE                                          | —                            | —                              |
| Cut-and-cover tunnel MDE                                          | —                            | —                              |
| Minimum published tunnel emergency airflow                        | —                            | —                              |
| Maximum published tunnel emergency airflow                        | —                            | —                              |
| Simulated train fire-resistance time in the Circular design study | —                            | —                              |
| Circular Line vehicle intercar gangway width                      | —                            | —                              |

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Station names, codes and sequence from TDX Transport Data eXchange, Ministry of Transportation and Communications, operators TRTC, NTMC, TYMC, and TMRT. Government open data. Retrieved 2026-08-24; source last updated 2016-11-23. The 12 Sanying station records absent from TDX are sourced directly to New Taipei Metro and government publications on their pages. Provenance.